

ARUN KUMAR A R

Unity Developer | Game Development (AR / VR / XR) | 3D Interactive Experiences
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PROFESSIONAL SUMMARY

Results-driven Unity Developer with **4.9 years of experience** specialising in **Game Development, Augmented Reality (AR), Virtual Reality (VR), and Extended Reality (XR)** across enterprise and consumer applications. Proven expertise in **Unity 3D, C# programming, Object-Oriented Programming (OOP), performance optimization**, and cross-platform deployment on **Android, iOS, PC (Windows), WebGL**, and **Oculus/Meta Quest** platforms. Successfully delivered immersive interactive experiences for high-profile clients including **Honeywell, KidZania**, and **MetaGP**. Experienced in **API integration, RESTful backend connectivity, and TCP/IP networking**. Currently expanding knowledge in **multiplayer systems and networking frameworks (Photon, Netick 2, Unity Netcode)**. Passionate about pushing the boundaries of real-time 3D, interactive gameplay mechanics, and XR applications.

TECHNICAL SKILLS

Game Engine & IDE:	Unity 3D (2019 / 2021 / 2022 LTS), Visual Studio, Scene Management, Game Logic Programming and UE5 Basic Scene Setup
Programming:	C# (Primary), Object-Oriented Programming (OOP), SOLID Principles, JSON Serialisation, Shader Graph, JavaScript, HTML, CSS
AR / VR / XR:	AR Foundation, ARCore, ARKit, Oculus SDK, Meta Quest 2/Pro, XR Interaction Toolkit (XRIT), OpenXR, SteamVR and Extended Reality (XR)
Rendering & Graphics:	URP, HDRP, Built-in Pipeline, Shader Graph, VFX Graph, Post-Processing Stack, Real-Time & Baked Lighting
Performance Optim.:	CPU & GPU Optimisation, Draw Call Batching & Reduction, LOD Systems, Occlusion Culling, Memory Management, Unity Profiler
Gameplay & Physics:	Gameplay Mechanics Implementation, Unity Physics, Animator Controller, Blend Trees, Particle Systems, UI Systems (Canvas, TextMeshPro)
Networking & Backend:	API Integration, RESTful APIs, TCP/IP Networking, JWT Authentication; actively learning Multiplayer Networking (Photon, Netick 2 ,Unity Netcode)
Async & Architecture:	Delegates, Events, Coroutines, Addressable Asset System, Data Management
3D & Creative Tools:	Blender (3D Modelling & Optimisation), SpeedTree, MeshRoom (Photogrammetry), DaVinci Studio, Audacity, PhotoPea
Platforms:	Android, iOS, PC (Windows Standalone), WebGL, Mac, Oculus/Meta Quest, iPad OS
Version Control:	Git
AI Tools:	Claude, Gemini, Flow, Suno AI

PROFESSIONAL EXPERIENCE

Unity Developer — Game Development, 3D & VR/XR

Tiltlabs Consultancy Services | Trivandrum, Kerala, India | February 2022 – Present

- Engineered comprehensive VR asset bundles, realistic plant animations, and interactive equipment simulations for the **Honeywell VR Training Application**, improving employee training efficiency by over 40% in petroleum plant operations.
- Developed the immersive **MetaGP Garage VR Experience** featuring first-person character navigation, JWT Token authentication, Addressable Asset System for dynamic asset loading, and cross-platform VR optimisation; achieved consistent **72+ FPS on Meta Quest 2** through draw call reduction, LOD, and URP tuning.
- Spearheaded **KidZania TVS Bike Studio** — drag-and-drop bike customisation kiosk, automated certificate generation with video recording, QR code creation, and **AWS S3** integration; reliably supporting 100+ daily sessions.
- Designed core gameplay mechanics for **KidZania Mahindra Lifespaces** township kiosk including draggable building systems, dynamic scoring, PNG certificate printing, and GPU-optimised scene lighting with post-processing.
- Built **GenRobotics VR Experience** on Oculus Quest using **TCP socket-based real-time video streaming**, delivering a fully networked VR video playback system controlled via a remote socket server.
- Delivered **Serenity PathWays VR** cognitive training app featuring physics-based puzzles, progressive difficulty levels, motor skill mechanics, and immersive spatial audio for clinical therapeutic use.

- Created **Emcode Union Boat** kayak racing simulation integrating real-world motion data for realistic paddling physics, in-game distance tracking, and responsive UI/UX.
- Developed **Honeywell AR Surface Scan App** using AR Foundation for iPad OS — dynamic UI labelling, 3D model outlines, interactive scaling, and cross-section views.
- Delivered **Cricket VR Arcade Ticketing App** with Bluetooth printer connectivity, real-time validation, and ticket ID generation.

Unity Developer — 3D, VR & Game Development

Embright Infotech Pvt Ltd | Trivandrum, Kerala, India | March 2021 – February 2022

- Developed **VR Laser and Light Reflection Simulation** for Oculus Quest using Unity Physics to accurately simulate laser behaviour across mirrors, wood, and metal surfaces.
- Optimised interior lighting and scene setup for **ULCC Archviz** architectural visualisation using baked lighting for photorealistic walkthrough performance.
- Built specialised VR scenarios and interactive learning modules for **Auticare VR**, an educational platform for children with Autism Spectrum Disorder (ASD).
- Engineered **Alphabet Learning** mobile application featuring drag-and-drop educational gameplay, 3D asset touch-rotation interactions, and structured app management for child learning.

KEY PROJECTS

Honeywell AR Surface Scan App | Lead Unity Developer | 2023

Tech Stack: Unity 3D, AR Foundation, ARCore, ARKit, C#, URP, iPad OS

- Developed end-to-end AR application enabling field engineers to visualise predefined 3D models on physical surfaces with interactive cross-section views, real-time scaling, and multi-touch controls.
- Implemented dynamic UI label system and model part outlines, significantly improving field engineer workflow efficiency and on-site decision-making.

MetaGP Garage VR Experience | Unity Developer | 2022–2023

Tech Stack: Unity 3D, C#, Oculus SDK, Meta Quest 2, Addressable Asset System, JWT, URP

- Built immersive first-person VR motorsport garage experience with secure user authentication, dynamic asset loading, and voice-interactive 3D information cards.
- Optimised cross-platform VR performance achieving consistent **72+ FPS on Meta Quest 2** through draw call reduction, LOD implementation, and URP tuning.

KidZania TVS Bike Studio | Unity Developer | 2022

Tech Stack: Unity 3D, C#, AWS S3, QR Code Integration, Video Recording, URP

- Engineered interactive bike customisation kiosk with drag-and-drop gameplay mechanics, real-time 3D preview, automated certificate generation pipeline, QR code creation, and secure AWS S3 upload supporting 100+ daily sessions.

Serenity PathWays VR | Unity Developer | 2023

Tech Stack: Unity 3D, C#, Oculus SDK, Meta Quest, Unity Physics, URP, Spatial Audio

- Developed VR cognitive training application with physics-based interactive puzzles, progressive difficulty system, and spatial audio feedback for clinical therapeutic environments.

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science & Engineering

College of Engineering, Kottarakara | Kottarakara, Kerala, India | 2016 – 2020 | CGPA: 7.4 / 10